Genni: Automated Radiology Report Scribe

Project Analysis Report

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Contents

Repository Understanding Summary

The project, **Automated Radiology Report Scribe**, aims to generate preliminary radiology-style narratives from medical images, focused on MRI head images related to tumors, cancer, carcinoma, metastasis, mass, and similar findings. The clinical objective is workflow support: reducing manual narrative drafting burden while describing visible abnormalities in MRI scans.

The dataset is the **MultiCaRe MRI subset**, comprising **1,703 MRI head image-caption pairs** from **831 case IDs**, converted into a LLaVA-style conversational JSON dataset. The pipeline uses `Salesforce/blip-image-captioning-base` as the baseline vision-language model (VLM), then fine-tunes BLIP on the project dataset. Evaluation compares zero-shot BLIP against the fine-tuned checkpoint using BLEU-1 through BLEU-4 and ROUGE-1/2/L over all 1,703 generated captions.

Key Experimental Findings

Fine-tuning substantially improved text overlap with ground-truth captions, as summarised in Table 1.

Table 1: Quantitative comparison: Zero-shot BLIP vs. Fine-tuned BLIP

Metric	**Zero-Shot**	**Fine-Tuned**
BLEU-1	0.0350	0.1826
BLEU-avg	0.0100	0.0611
BLEU-4	0.0048	0.0228
ROUGE-1 F	0.0878	0.2726
ROUGE-2 F	0.0073	0.0793
ROUGE-L F	0.0716	0.2013

The zero-shot model failed badly for this clinical domain. Approximately **57.4%** of zero-shot outputs contained heavy repeated "mri" tokens, and some outputs described irrelevant natural-image content such as dogs, cats, knees, or a telescope-like image.

The fine-tuned model learned the form and vocabulary of radiology captions: all fine-tuned outputs contained medical/imaging terminology by simple keyword audit, and token-level collapse disappeared. However, the absolute scores remain low, especially **BLEU-4 = 0.0228**, and many outputs are template-like. A keyword audit found fabricated or malformed medical-like terms in approximately **45.6%** of `fine_tuned_results.txt`, including fragments such as "edemastrens," "ventriclebral," and "cranistic." This constitutes the central clinical safety limitation.

Analysis

Overview of Experimental Outcomes

The project successfully demonstrated that domain adaptation improves medical image caption generation compared with an off-the-shelf vision-language baseline. The zero-shot BLIP model was not clinically meaningful on the MultiCaRe MRI subset: it frequently collapsed into repetitive tokens or produced generic natural-image descriptions unrelated to MRI interpretation. After fine-tuning on the 1,703 MRI image-caption pairs, the model generated fluent radiology-style captions containing relevant terms such as MRI, T1-weighted imaging, mass, lesion, tumor, ventricle, cerebellum, contrast, and preoperative/postoperative descriptors.

Quantitatively, the fine-tuned model improved across every reported BLEU and ROUGE metric. BLEU-1 increased from 0.0350 to 0.1826, BLEU-avg from 0.0100 to 0.0611, ROUGE-1 F from 0.0878 to 0.2726, and ROUGE-L F from 0.0716 to 0.2013. These gains indicate that fine-tuning helped the model learn the vocabulary and sentence structure of the target radiology-caption distribution. However, the absolute scores remain modest, especially BLEU-4 at 0.0228, showing that the model still rarely matches longer clinically specific phrases from the references.

What Worked Well

The strongest result was the clear improvement from zero-shot BLIP to the fine-tuned model. Fine-tuning eliminated the most severe zero-shot failure mode: repeated "mri" or "ct" tokens. It also shifted outputs from generic descriptions toward radiology-style narratives. The generated captions after fine-tuning usually follow a recognisable clinical-caption pattern: modality or sequence first, then anatomical location, then a described abnormality or postoperative finding. This suggests that the model learned a useful surface-level mapping from the MultiCaRe caption distribution.

The training run also appears computationally successful under constrained hardware. The full-training manifest reports completion of **10 epochs** with batch size 1, gradient accumulation 8, learning rate 2×10^{-5}, and effective batch size 8. Loss decreased from **8.3820** at epoch 1 to **2.0123** at epoch 10, with the steepest improvement in the first five epochs. The absence of recorded failed inference cases is also positive: the fine-tuned inference manifest reports 1,703 successful generations and zero failures.

What Failed or Underperformed

The major failure is that improved lexical overlap did not translate into reliable clinical accuracy. The fine-tuned captions often sound medically plausible while including incorrect or nonsensical terminology. Examples in the generated outputs include malformed terms such as "cranistic," "edemastrens," "ventriclebral," and "ventriclectomial." This means the model learned radiology-like language patterns but not enough grounded visual-clinical reasoning to be trusted.

The fine-tuned model also shows strong **template behaviour**. Many outputs begin with similar constructions, especially "preoperative t1-weighted magnetic resonance imaging. . . " and repeatedly mention a small set of structures such as right parietal lesion, left temporal lobe, ventricle, and cerebellum. This repetition suggests the model may be overusing common caption priors rather than discriminating precisely between individual MRI images.

Evaluation is another limitation. The reported metrics are computed over all 1,703 samples, and the available metric artefacts do not provide a clean held-out test-set score. The current training code contains validation-split support, but the available `full_training_manifest.json` has 213 optimiser steps per epoch, consistent with using all 1,703 samples with gradient accumulation 8, and it does not contain validation-loss fields. Therefore, the strongest supported claim is full-dataset evaluation, not independent generalisation.

Comparison Between Models

The zero-shot BLIP baseline was not suitable for medical MRI captioning. Its BLEU-4 score of 0.0048 and ROUGE-2 F score of 0.0073 reflect almost no meaningful phrase-level overlap with the clinical references. Qualitatively, it often generated repetitive outputs or irrelevant natural-image captions.

The fine-tuned BLIP model was substantially better on every reported metric. It achieved BLEU-1 of 0.1826, BLEU-4 of 0.0228, ROUGE-1 F of 0.2726, ROUGE-2 F of 0.0793, and ROUGE-L F of 0.2013. This shows successful domain adaptation at the vocabulary and caption-structure level. Still, the model remains far from clinically reliable because it hallucinates medical language and produces repetitive, template-driven findings.

Clinical Relevance and Impact

Clinically, the project is best interpreted as an **early prototype** for radiology narrative assistance, not as a deployable diagnostic system. The fine-tuned model could potentially support workflow exploration by producing draft-style captions or demonstrating how VLMs adapt to MRI reporting language. It may reduce some low-level drafting burden in a tightly supervised setting.

However, the current model should **not** be used for clinical decision-making. False or hallucinated descriptions of lesion location, tumour presence, sequence type, or postoperative status could mislead clinicians if treated as factual. The most clinically important risk is not that the model fails obviously, but that it *fails fluently*: it produces plausible report-like language that may be anatomically or pathologically wrong. Any real-world use would require radiologist review, held-out clinical validation, hallucination controls, and evaluation metrics that measure factual correctness rather than only text overlap.

Limitations and Future Improvements

Future work should first establish a strict train/validation/test split at the patient or case level to avoid evaluating on training-distribution examples. The project should also add clinically meaningful evaluation: radiologist scoring, error categorisation, anatomy/location correctness, finding presence/absence accuracy, and hallucination rate. BLEU and ROUGE are useful for ablation, but they are not sufficient for patient-safety claims.

Model improvements should focus on grounding and reliability. The following directions are recommended:

- Retrieval-augmented captioning
- Stronger medical VLM backbones
- Parameter-efficient fine-tuning (e.g. LoRA)
- Image augmentation
- Constrained clinical vocabulary generation

Larger and more diverse MRI datasets would also be needed before claiming generalisability beyond this MultiCaRe subset.

Evidence Traceability

The analysis is grounded in the following repository artefacts:

1. **Domain Research note.pdf** — project objective as automated radiology narrative generation.
2. **readme.txt** (`vlm_mri_subset/`) — MultiCaRe subset filters and dataset source.
3. **prepare_vlm_dataset.py** — conversion to conversational image-caption format.
4. **train_full.py** — BLIP fine-tuning setup and hyperparameters.
5. **full_training_manifest.json** — 10 completed epochs and loss trajectory.
6. **metrics_zero_shot.json** and **metrics_fine_tuned.json** — quantitative comparison.
7. **preliminary_results.txt** and **fine_tuned_results.txt** — qualitative output behaviour.
8. **inference_manifest.json** — 1,703/1,703 successful fine-tuned inference runs.

Assumptions and Caveats

The actual checkpoints, model cache, logs, and SQLite evaluation database are not present in the repository; they are referenced but gitignored or absent. Additionally, `experiments_and_results.md` describes a 90/10 validation split, but the available result artefacts support full-dataset metric evaluation on 1,703 samples and do not include held-out validation/test metrics. The result files were treated as the authoritative evidence throughout this report.